

The modified fpca.face function in refund

Ruonan Li

1. Algorithm.

For subject $i = 1, \dots, I$ denote by $Y_i(t)$ the observed data at $t \in \mathcal{T}$. Assume that data are observed on a regular grid of the domain, that is $\{t_1, \dots, t_J\}$, where J is the total number of grid points. Define $\mathbf{Y}$ the demeaned observed functional data, which is a $J \times I$ dimensional matrix with the i th column equal to $\mathbf{Y}_i$.

Part 1: FACE algorithm in Xiao et al. (2016).

- Step 1: conduct the eigendecomposition $(\mathbf{B}^T \mathbf{B})^{-1/2} \mathbf{P} (\mathbf{B}^T \mathbf{B})^{-1/2} = \mathbf{U} \mathbf{D} \mathbf{U}^T$, where $\mathbf{U}$ is an orthonormal matrix and $\mathbf{D}$ is a diagonal matrix.
- Step 2: calculate $\mathbf{A}_S = \mathbf{B} (\mathbf{B}^T \mathbf{B})^{-1/2} \mathbf{U}$.
- Step 3: calculate $\tilde{\mathbf{Y}} = \mathbf{A}_S^T \mathbf{Y}$.
- Step 4: conduct the eigendecomposition $I^{-1} (\mathbf{I}_c + \lambda \mathbf{D})^{-1} \tilde{\mathbf{Y}} \tilde{\mathbf{Y}}^T (\mathbf{I}_c + \lambda \mathbf{D})^{-1} = \mathbf{A} \mathbf{\Sigma} \mathbf{A}^T$, where $\mathbf{A}$ is an orthonormal matrix and $\mathbf{\Sigma}$ is a diagonal matrix.
- Step 5: calculate eigenvectors of $\tilde{\mathbf{K}}$ as $\mathbf{A}_S \mathbf{A}$.

Part 2: The modified FACE algorithm.

- Step 1: conduct the eigendecomposition $(\mathbf{B}^T \mathbf{B})^{-1/2} \mathbf{P} (\mathbf{B}^T \mathbf{B})^{-1/2} = \mathbf{U} \mathbf{D} \mathbf{U}^T$, where $\mathbf{U}$ is an orthonormal matrix and $\mathbf{D}$ is a diagonal matrix.
- Step 2: calculate $\mathbf{A}_0 = (\mathbf{B}^T \mathbf{B})^{-1/2} \mathbf{U}$ and $\mathbf{A}_S = \mathbf{B} (\mathbf{B}^T \mathbf{B})^{-1/2} \mathbf{U}$.
- Step 3: calculate $\tilde{\mathbf{Y}} = \mathbf{A}_S^T \mathbf{Y}$.
- Step 4: construct transformed basis functions as $\tilde{\mathbf{B}}(s)$ so that they are orthonormal, i.e., $\int \tilde{\mathbf{B}}(s) \tilde{\mathbf{B}}^T(s) ds = \mathbf{I}_c$. We have $\tilde{\mathbf{B}} = \mathbf{B} \mathbf{G}^{-1/2}$ and $\tilde{\mathbf{A}}_0 = \mathbf{G}^{1/2} \mathbf{A}_0$, where $\mathbf{G} = \int \mathbf{B}(s) \mathbf{B}^T(s) ds$.
- Step 5: conduct the eigendecomposition $\tilde{\mathbf{A}}_0 \{n^{-1} (\mathbf{I}_c + \lambda \mathbf{D})^{-1} \tilde{\mathbf{Y}} \tilde{\mathbf{Y}}^T (\mathbf{I}_c + \lambda \mathbf{D})^{-1}\} \tilde{\mathbf{A}}_0^T = \mathbf{A} \mathbf{\Sigma} \mathbf{A}^T$, where $\mathbf{A}$ is an orthonormal matrix and $\mathbf{\Sigma}$ is a diagonal matrix.
- Step 6: The estimated k th eigenfunction is $\phi_k(s) = \mathbf{A}_k^T \tilde{\mathbf{B}}(s)$, where $\mathbf{A}_k$ is the k th column vector of $\mathbf{A}$.

We use mixed model equations (MME) to estimate principal scores. Assume that we obtained $Y_i(t)$, eigenvalues λ_k , and eigenfunctions $\phi_k(t)$, $k = 1, \dots, N$ from previous steps. N is selected using the proportion of explained variance. The mixed model is

$$\begin{cases} Y_i(t) = \sum_{k=1}^N \xi_{ik} \phi_k(t) + \epsilon_i(t), \\ \xi_{ik} \sim \mathcal{N}\{0, \lambda_k\}; \quad \epsilon_i(t) \sim \mathcal{N}(0, \sigma^2). \end{cases} \quad (1)$$

Data may be incompletely collected, that is $\{t_1, \dots, t_{T_i}\}$. Let $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{iT_i})^T$ and $\boldsymbol{\xi}_i = (\xi_{i1}, \dots, \xi_{iN})^T$. Note that the score $\boldsymbol{\xi}_i$ has a diagonal covariance matrix $\mathbf{\Sigma} = \text{diag}(\lambda_1, \dots, \lambda_N)$. Let $\boldsymbol{\phi}_{k,i} = (\phi_k(t_1), \dots, \phi_k(t_{T_i}))^T$ and $\boldsymbol{\Phi}_i = [\boldsymbol{\phi}_{1,i}, \dots, \boldsymbol{\phi}_{N,i}] \in \mathbb{R}^{T_i \times N}$. Let $\boldsymbol{\epsilon}_i = (\epsilon_{i1}, \dots, \epsilon_{iT_i})^T$. Then

$$\mathbf{Y}_i = \boldsymbol{\Phi}_i \boldsymbol{\xi}_i + \boldsymbol{\epsilon}_i.$$

By MME, we have

$$\boldsymbol{\xi}_i = \{\boldsymbol{\Phi}_i^T \boldsymbol{\Phi}_i + \sigma^2 \mathbf{\Sigma}^{-1}\}^{-1} \boldsymbol{\Phi}_i^T \mathbf{Y}_i, \quad (2)$$

where the dimension of the matrix to take inverse is $N \times N$. Then, $\hat{\mathbf{Y}}_i = \mathbf{\Phi}_i \boldsymbol{\xi}_i$.

2. Simulation: compare the computation time and estimate accuracy of `fpca.face` and `fpca.face2`.

2.1 Equally-spaced data.

We assume that the regular functional data are observed on an equally-spaced grid $\{t_1, \dots, t_J\}$ of domain $\mathcal{T} = [0, 1]$ such that $t_j = j/J$ for $j = 1, \dots, J$. We also consider an incomplete scenario, where the number of observed points per function is $0.5 \times J$. We considered $I = 100, 200, 500$ subjects, and $J = 100, 1000, 10000$ points per curve. The magnitude of noise $\sigma = 0.5$.

We use a similar simulation setting with Xiao et al. (2016). Consider the following model

$$Y_i(t) = \sum_{k=1}^3 \xi_{ik} \phi_k(t) + \epsilon_i(t),$$

where $\xi_{ik} \sim \mathcal{N}\{0, \lambda_k\}$ and $\epsilon_i(t) \sim \mathcal{N}(0, \sigma^2)$. The true eigenvalues are $\lambda_k = 0.5^{k-1}, k = 1, 2, 3$. The true eigenfunctions are selected as $\phi_k(t) = \{1, \sqrt{3}(2t - 1), \sqrt{5}(6t^2 - 6t + 1)\}$. Those eigenfunctions are not mutually orthogonal.

I simulated 100 data sets for each scenario and compared `fpca.face` and `fpca.face2` by computation time, mean integrated square errors (MISE) for eigenfunctions and mean square errors (MSE) for estimated scores. Results are shown in Table 1.

Table 1: Equally-spaced data: the average MISE for eigenfunctions and the average MSE for scores across 100 simulations.

deisgn	J	n	fpca.face			fpca.face2		
			time (s)	$MISE(\phi)$	MSE	time (s)	$MISE(\phi)$	MSE
regular	100	100	0.0110	0.0329	0.0252	0.0268	0.0310	0.0241
		200	0.0070	0.0155	0.0133	0.0175	0.0143	0.0126
		500	0.0109	0.0070	0.0069	0.0341	0.0063	0.0066
	1000	100	0.0171	0.0336	0.0235	0.0731	0.0334	0.0234
		200	0.0269	0.0143	0.0109	0.0951	0.0142	0.0109
		500	0.0884	0.0061	0.0046	0.1921	0.0061	0.0046
	10000	100	0.2120	0.0341	0.0234	0.4492	0.0341	0.0234
		200	0.3607	0.0144	0.0108	0.4634	0.0144	0.0108
		500	0.7213	0.0060	0.0045	0.9829	0.0060	0.0045
irregular	100	100	0.0386	0.0318	0.0268	0.0686	0.0301	0.0258
		200	0.0740	0.0155	0.0155	0.1025	0.0144	0.0149
		500	0.1717	0.0070	0.0093	0.2310	0.0064	0.0090
	1000	100	0.1197	0.0325	0.0231	0.1479	0.0324	0.0230
		200	0.2251	0.0141	0.0110	0.2694	0.0141	0.0109
		500	0.5618	0.0062	0.0048	0.6467	0.0062	0.0048
	10000	100	0.8898	0.0339	0.0233	0.9990	0.0339	0.0233
		200	1.6231	0.0144	0.0108	1.8213	0.0144	0.0108
		500	5.9102	0.0061	0.0045	7.4222	0.0061	0.0045

2.2 Unequally-spaced data.

For unequally-spaced data, data are also observed on J points. The J points are randomly sampled from an equally-spaced grid $\{t_1, \dots, t_{5J}\}$ of domain $\mathcal{T} = [0, 1]$. Results are shown in Table 2.

Table 2: Unequally-spaced data: the average MISE for eigenfunctions and the average MSE for scores across 100 simulations.

deisgn	J	n	fpca.face			fpca.face2		
			time (s)	$MISE(\phi)$	MSE	time (s)	$MISE(\phi)$	MSE
regular	100	100	0.0076	0.0717	0.0418	0.0154	0.0379	0.0274
		200	0.0064	0.0464	0.0257	0.0167	0.0132	0.0119
		500	0.0092	0.0385	0.0200	0.0266	0.0066	0.0068
	1000	100	0.0159	0.0391	0.0260	0.0353	0.0374	0.0259
		200	0.0376	0.0194	0.0137	0.0560	0.0148	0.0116
		500	0.0693	0.0102	0.0065	0.1590	0.0064	0.0049
	10000	100	0.1750	0.0296	0.0214	0.2213	0.0292	0.0213
		200	0.2334	0.0172	0.0120	0.2746	0.0165	0.0118
		500	0.4932	0.0064	0.0048	0.6069	0.0061	0.0047
irregular	100	100	0.0468	0.0729	0.0435	0.0586	0.0364	0.0288
		200	0.0800	0.0473	0.0280	0.1015	0.0134	0.0144
		500	0.1847	0.0387	0.0221	0.2380	0.0068	0.0093
	1000	100	0.1219	0.0386	0.0259	0.1506	0.0368	0.0257
		200	0.2243	0.0192	0.0137	0.2750	0.0147	0.0116
		500	0.5765	0.0103	0.0066	0.6458	0.0065	0.0051
	10000	100	0.8138	0.0295	0.0213	0.8874	0.0290	0.0212
		200	1.5079	0.0171	0.0120	1.7151	0.0164	0.0118
		500	5.1945	0.0064	0.0048	6.5421	0.0061	0.0047